

# AI for Education Initiative

Building Local AI Infrastructure for Teachers and Students

Prepared by William Morris | May 2026

## Executive Summary

We propose establishing a dedicated AI education server at NAS Jiaxing — a local, privacy-focused platform that gives teachers and students direct access to cutting-edge AI tools without relying on external cloud services or VPNs.

### The Problem We Are Solving

Teachers want to use AI in classrooms, but face three barriers:

- Access blocked** — Many AI tools (ChatGPT, Claude, Gemini) are blocked on school networks without VPN
- Data privacy** — Uploading student work to foreign cloud services raises compliance concerns
- Cost and complexity** — Premium AI subscriptions are expensive; teachers don't know where to start

## Our Solution

A local AI server running inside the school network, hosting open-source AI models and educational tools that are:

- Always accessible** — No VPN needed; works on school WiFi
- Fully private** — All data stays inside the school; no external cloud uploads
- Free to use** — Open-source software; no subscription fees
- Educational** — Designed specifically for teaching, not general chat

## What Is Already Working (Demo)

We have built a proof-of-concept running on a personal workstation. This demonstrates that the technology works and the concept is viable.

### Current Demo Machine Specs

| Component | Specification                              |
|-----------|--------------------------------------------|
| CPU       | AMD Ryzen 9 7950X3D (16 cores, 32 threads) |

|                  |                                            |
|------------------|--------------------------------------------|
| RAM              | 96GB DDR5                                  |
| GPU              | AMD RX 7900 XTX (24GB VRAM)                |
| Storage          | 2TB NVMe SSD + 4TB HDD                     |
| OS               | Ubuntu 24.04 LTS                           |
| Network          | School LAN — accessible from any classroom |
| Total Build Cost | ~¥20,000 (personal funds)                  |

This machine currently serves educational content and AI tools to the school network. It proves the concept. For a permanent school solution, we recommend a purpose-built server at roughly half the cost.

# Proposed School Server

For a permanent, school-owned installation, we recommend a dedicated server that is:

- More reliable (enterprise-grade components)
- More maintainable (remote management, warranty support)
- More cost-effective (target ~¥10,000 vs. the ¥20,000 demo build)
- Physically secured in the school server room

## Recommended Server Specs (~¥10,000)

| Component            | Specification                                              | Est. Price |
|----------------------|------------------------------------------------------------|------------|
| CPU                  | AMD Ryzen 7 7700X (8 cores, 16 threads) or used server CPU | ¥2,000     |
| RAM                  | 64GB DDR5 (expandable to 128GB)                            | ¥2,500     |
| GPU                  | AMD RX 7800 XT (16GB VRAM) or used compute GPU             | ¥3,500     |
| Storage              | 1TB NVMe SSD + 4TB HDD                                     | ¥1,200     |
| Case/PSU/Motherboard | Standard ATX or used server chassis                        | ¥1,500     |
| OS & Software        | Ubuntu 24.04 LTS (free, open-source)                       | ¥0         |
| Total                |                                                            | ~¥10,200   |

Alternative: A used enterprise server (e.g., Dell PowerEdge or HP ProLiant) from second-hand market can provide similar specs at ¥8,000–12,000 with better reliability and remote management (iDRAC/iLO).

# What the Server Will Host

## For Teachers — Professional Development

- **AI Lesson Plan Assistant** — Generate differentiated activities, rubrics, and assessments
- **Text Simplifier** — Adapt reading passages for ESL students by level
- **Discussion Starter Generator** — Create engaging prompts for any topic
- **Vocabulary Builder** — Auto-generate word lists with definitions and example sentences
- **Reading Assessment Creator** — Build comprehension questions with answer keys

## For Students — Learning Tools

- **Educational Games** — Grammar, vocabulary, and critical thinking games
- **Creative Project Generator** — AI-assisted story writing and project ideas
- **AI Tutor** — Students can ask questions and get explanations without leaving the school network
- **Listening Practice** — AI-generated listening exercises with transcripts

## For the School — Infrastructure

- **Internal AI Resource Website** — [ai.nasjiaxing.cn](http://ai.nasjiaxing.cn) — a professional portal for all AI tools
- **Content Management** — IT controls what content is available; everything is curated for education
- **No External Dependencies** — Works even if the school's internet connection is slow or filtered

# Benefits to NAS Jiaxing

## For Teachers

- Save hours on lesson planning and material creation
- Differentiate instruction automatically
- Access PD workshop materials anytime
- No need for personal VPNs or subscriptions
- Data stays inside the school — fully compliant

## For Students

- Personalized learning tools available 24/7 on campus
- Interactive games that reinforce curriculum
- AI writing and creativity support
- Safe environment — no ads, no tracking, no external data sharing

## For the School

- Position NAS Jiaxing as a leader in AI education
- One-time hardware cost vs. ongoing subscription fees
- Full control over content and access
- Scalable — add more capabilities as needed

## For IT & Compliance

- All traffic stays on the school LAN
- No student data leaves the building
- Can be placed under school's existing ICP备案
- Centrally managed and monitored

# Implementation Timeline

| Phase                           | Duration  | Actions                                                           |
|---------------------------------|-----------|-------------------------------------------------------------------|
| Phase 1: Approval & Procurement | 2–3 weeks | Principal and IT approve budget; order server hardware            |
| Phase 2: Setup & Configuration  | 1 week    | Install Ubuntu, configure AI software, set up internal website    |
| Phase 3: Content & Testing      | 1–2 weeks | Load educational tools, test with pilot teachers, gather feedback |
| Phase 4: Launch & Training      | 1 week    | Internal launch, PD workshop for staff, onboarding materials      |
| Phase 5: Expansion              | Ongoing   | Add new tools based on teacher requests, student feedback         |

# Compliance & Security

We are committed to full compliance with Chinese regulations:

- **ICP备案** — The site will operate under the school's existing ICP filing as a subdomain (ai.nasjiaxing.cn)
- **Data Sovereignty** — All data stored locally; nothing sent to foreign servers
- **Content Control** — IT and school leadership review and approve all content before go-live
- **No VPN Required** — Everything runs on the school internal network
- **Firewall Friendly** — Standard HTTP on port 80; no special ports or protocols needed

# What We Need From the School

## Three Requests to Move Forward

### 1. Budget Approval (~¥10,000)

For a dedicated school AI server. We can source used enterprise hardware to stay within budget.

### 2. IT Support — Static IP & Subdomain

A reserved static IP on the school LAN for the server, and a DNS entry for ai.nasjiaxing.cn pointing to that IP.

### 3. ICP Subdomain Authorization

Permission to add ai.nasjiaxing.cn under the school's existing ICP备案.

# About the Demo

A live demo is already running on the school network. You can access it from any device connected to the school WiFi at:

http://10.39.26.217/

Accessible only from the school LAN

This demo shows the landing page, educational games, AI tools, and the general look and feel of what the final site would be.

# About Me — William Morris

I am an English teacher at NAS Jiaxing with a background in technology and a personal interest in AI. I built the demo machine in my own time using personal funds to prove that this is feasible and valuable for our school. I am not asking the school to fund my personal project — I am proposing a separate, school-owned server that serves the entire faculty and student body.

I am happy to manage the technical setup, maintain the server, and train teachers on how to use the tools. My goal is to make NAS Jiaxing a model school for AI-integrated education.

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**Questions?** Contact William Morris at [william.morris@nasjiaxing.cn](mailto:william.morris@nasjiaxing.cn)

Full technical details available upon request · Demo accessible on school network